Time Limit	50 test inputs must be executed within 1 second(s) using C/C++, or 1.5 second(s) using Java
Memory Limit	Stack : 1 Mbytes / Total : 256 Mbytes
Code Length Limit	128 Kbytes
Evaluation	When you submit an answer, the result will be shown in real time as below. **"Pass" means that the answer is correct for all of the rest inputs.** **"Fail" means that the answer is incorrect for all of part of the test inputs**

You are given a grid of size NxM and each cell of the grid is filled with a number.

In the grid you want to color some of the cells by following the condition below.

> Condition: **By either adding or subtracting all of the numbers that are placed at the top, at the bottom, to the right and to the left of a given number**, you can get the given number

See the grid below. You are given a 5x6 grid and you can color a total of 18 cells in this grid.

3	4	0	-1	9	-2
-1	1	3	8	10	13
5	5	2	0	6	-4
-5	7	4	-1	8	-9
3	-7	-3	5	-3	12



- Example 1) 3 = (4) + (-1)
- Example 2) 10 = (9) + (8) + (6) - (13)
- Example 3) 5 = - (-3) + (-1) - (-3)

When numbers written in the cells are given as input, print how many cells you can color in that grid.

[Constraints]

1. The size of the grid N and M are integers (5 ≤ N, M ≤ 50).
2. The number given to each cell is an integer (-20 ≤ number ≤ 20).

[Input]

The first line contains T denoting the number of test cases. Then, T test cases follow.

The first line of each test case contains N and M, the size of a grid, separated by space.

The following N lines contain the numbers written in the grid cells. Each of the lines contains M integers, separated by space.

[Output]

For each test case, print #x (x is a test case number, starts from 1), Insert space, then print the number of the cells that can be colored.

[Output]

For each test case, print #x (x is a test case number, starts from 1), insert space, then print the number of the cells that can be colored.

[Example of Input and Output]

(Input)

```
5
5 6
3 4 0 -1 9 -2
-1 1 3 8 10 13
5 5 2 0 6 -4
-5 7 4 -1 8 -9
3 -7 -3 5 -3 12
9 5
-6 5 9 6 2
-7 -2 -5 0 7
2 6 10 -1 -1
6 -8 1 2 2
-6 -4 8 -10 8
9 -6 3 10 1
10 0 3 -2 5
-6 4 4 3 2
-6 10 8 -8 -5
7 12
-1 -4 3 1 -4 -6 4 -10 5 -8 -2 -3
5 1 10 -8 4 0 6 4 4 -5 -5 -10
-7 4 -7 9 10 -3 -10 -8 3 0 -5 1
-4 -8 -5 -6 -10 8 2 -1 -7 0 -9 -3
3 8 -5 5 -7 -5 8 -6 2 9 -3 4
10 -4 -6 2 8 -8 -9 6 6 -4 -10 9
0 -2 1 9 -8 -7 5 10 -6 -1 -8 -3
6 8
9 -7 0 0 9 4 2 -3
-6 0 -2 3 -1 -6 7 -5
-9 0 -4 6 3 -4 -2 -8
3 -8 5 -9 0 -1 9 8
-8 -10 -2 -6 1 -8 8 1
-4 -8 -1 7 -2 -7 6 10
5 9
-10 -10 -2 -3 10 9 -9 -8 0
9 -8 9 7 8 -3 -10 5 5
9 8 -5 10 -5 -7 -1 1 10
3 -4 -3 -3 -7 9 10 -5 2
9 10 -1 -9 7 5 3 -4 -2
```

(Output)

```
#1 18
#2 9
#3 22
#4 15
#5 19
```

(Output of sample_input.txt)

```
#1 18
#2 9
#3 22
#4 15
#5 19
#6 15
#7 11
#8 10
#9 81
#10 131
```

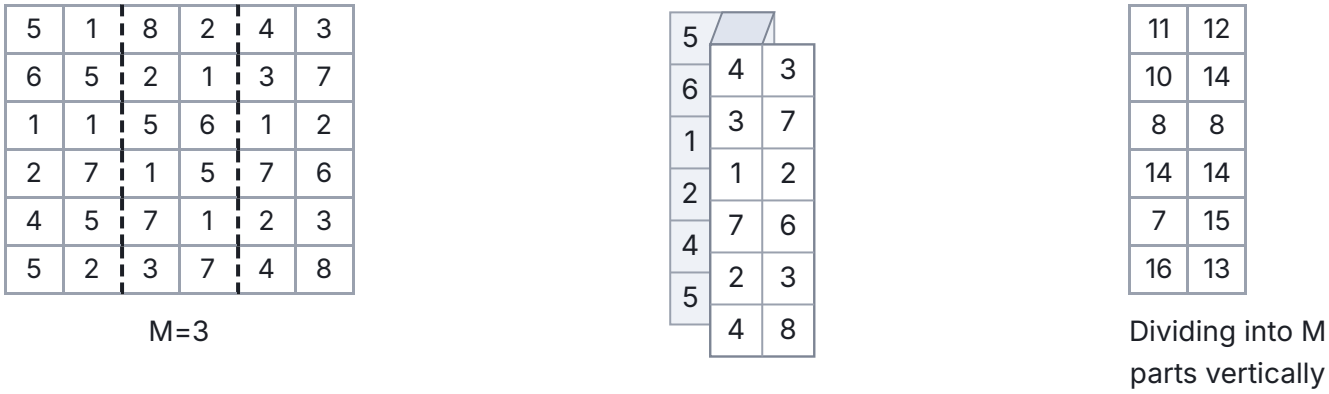
Problem: **(Level 1) [Official I-0019] Origami**

Time Limit	40 test inputs must be executed within 1 second using C/C++, or 1.5 seconds using Java
Memory Limit	Stack : 1 Mbytes / Total : 256 Mbytes
Maximum Number of Submission	10
Code Length Limit	128 KB
Evaluation	When you submit an answer, the result will be shown in real time as below. **"Pass" means that the answer is correct for all of the test inputs** **"Fail" means that the answer is incorrect for all of part of the test inputs** ※ Test Input : a data which is automatically inputted when submitted code is run in the test system

Minsu is doing origami. The paper for origami is N * N-sized square.

Minsu does stair fold in origami so that the size of paper stair should be N/M, meaning folding a paper in M equal parts.

Folding a paper repeatedly inward(to the left) and outward(to the right) is called stair fold. Let's find out the largest number of combining all the numbers overlapped on the paper.



[Figure]

The Figure is an example of paper whose N=6 and M=3. Add all the overlapped numbers up after folding the paper into M equal parts. In this case, the largest number is 16.

[Constraints]

1. N, representing Row and Column is an integer from 1 to 100, including 1 and 100.
2. M, the number of equal parts dividing the paper is an integer from 1 to N, including 1 and N.
3. Numbers on each of grids are integers from 1 to 100, including 1 and 100.
4. N is a number that multiplies M.

[Input]

The number of test cases T is given, followed by T test cases. N, the size of a paper and M, the number of equal parts of the paper are given as spaces in the first line of each case. The next N lines contains the information of the numbers on the paper which are separated by spaces..

[Output]

Print each test case to a separate line. Print #x (x=Test case number, beginning from 1), followed by a space and find the largest number.

[Input]

The number of test cases T is given, followed by T test cases. N, the size of a paper and M, the number of equal parts of the paper are given as spaces in the first line of each case. The next N lines contains the information of the numbers on the paper which are separated by spaces..

[Output]

Print each test case to a separate line. Print #x (x=Test case number, beginning from 1), followed by a space and find the largest number.

[Examples of Input and Output]

(Input)

```
2
6 3
5 1 8 2 4 3
6 5 2 1 3 7
1 1 5 6 1 2
2 7 1 5 7 6
4 5 7 1 2 3
5 2 3 7 4 8
8 4
3 4 8 7 5 1 6 4
5 4 7 8 1 3 2 4
4 8 6 7 2 1 6 8
2 1 5 7 6 4 8 2
3 4 8 6 9 1 2 3
5 5 4 7 6 2 8 4
4 9 8 7 1 8 6 5
1 2 4 8 7 9 5 3
```

(Output)

```
#1 16
#2 31
```

(Output of sample_input.txt)

```
#1 16
#2 31
#3 27
#4 33
#5 35
#6 902
#7 599
#8 506
#9 426
#10 1761
#11 575
#12 2021
#13 440
#14 5293
#15 1652
```